

SIH IDEA SUBMISSION REPORT

Low Cost Deployable Seafloor Metal Detection Sensor for Ocean Resource Exploration

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1. Introduction

1.1 Background

The deep ocean is one of the least explored regions on Earth and contains important mineral resources such as polymetallic nodules, polymetallic sulphides, cobalt rich crusts, and rare earth element bearing sediments. These resources are of growing interest worldwide due to their potential importance for advanced technologies and the increasing demand for critical minerals. However, exploring the deep seafloor is technically challenging because of extreme hydrostatic pressure, low temperature, darkness, corrosive seawater, and limited accessibility. Conventional surveys therefore rely on expensive research vessels, remotely operated vehicles (ROVs), autonomous underwater vehicles (AUVs), and specialized sensing systems.

The Indian Ocean is particularly significant in this context, with extensive regions containing mineral deposits of scientific and economic interest. India has undertaken systematic exploration of polymetallic nodules in the Central Indian Ocean Basin and polymetallic sulphides along the Indian Ocean ridges as part of its deep ocean research programmes. These activities require reliable and efficient technologies capable of surveying large and difficult to access seabed regions.

India's Deep Ocean Mission places strong emphasis on developing indigenous technologies for deep sea exploration, underwater robotics, and mineral resource assessment. In this context, low cost and deployable sensing systems can complement existing large scale survey platforms by enabling rapid preliminary assessment of metal rich regions. Developing such technology within India can also reduce dependence on imported instrumentation, simplify maintenance and customization, and strengthen domestic capabilities in underwater sensing, electronics, mechanical design, and ocean engineering.

The present project therefore focuses on developing an indigenous, low cost deployable seafloor metal detection system to support deep ocean resource exploration and seabed mapping.

1.2 Environmental & Physical Boundary Conditions (3,000 m – 6,000 m)

Designing instrumentation for abyssal exploration requires operating within extreme physical constants:

- **Hydrostatic Pressure:** Increases by approximately 1 bar (0.1 MPa) for every 10 meters of depth, reaching **30 MPa (300 bar / ~4,351 psi)** at 3,000 meters and **60 MPa (600 bar / ~8,702 psi)** at 6,000 meters. Unprotected electronic cavities suffer mechanical crushing and structural micro-fracturing.
- **Seawater Bulk Conductivity:** Abyssal seawater maintains a conductivity of **3.0 to 5.2 S/m** at salinities of 34.6 to 34.8 PSU, creating a continuous electromagnetic shield.
- **Signal Attenuation Limits:**
 - *Radio Frequencies:* Frequencies ≥ 1 MHz extinguish completely within **0.01 to 0.25 meters** of seawater, preventing any GPS, Wi-Fi, or radio links through the water column.
 - *Optical Extinction:* Zero sunlight reaches below 1,000 meters. Artificial optical visibility is restricted to **3 to 10 meters** even with high-lumen ($> 10,000$ lm) systems due to severe particulate backscatter.
- **Acoustic Channel Constraints:** Sound travels at **1,480 to 1,540 m/s**, causing a **~4.0-second** one-way transit delay over 6,000 meters. Vertical bandwidth is severely constrained to **80 bps to 15,360 bps** (0.08 to 15.3 kbps), restricting acoustic communication strictly to numeric status packets.
- **Thermal Regimes:** Ambient seafloor temperatures remain fixed at **1.0°C to 4.0°C** (specifically 1.2°C to 1.8°C in the Central Indian Ocean Basin).
- **Sediment Mechanics:** Abyssal plains consist of fluid pelagic oozes with an upper-layer shear strength of only **1.0 to 10.0 kPa** and a moisture content of **60% to 85% by weight**.

1.3 Core Engineering & Physics Implementation Challenges

- **Housing Selection & Electromagnetic Transparency:** At 60 MPa hydrostatic loads, metallic hulls induce parasitic eddy currents that distort magnetometer and EMI readings, while requiring dense syntactic foam for buoyancy.
- **Low-Temperature Battery Derating:** Ambient seafloor temperatures of 1.0°C to 4.0°C increase internal cell impedance by **2× to 4×** and reduce effective Lithium-ion capacity by **20% to 35%**, requiring ultra-low-power sleep cycles and strict thermal design margins.
- **Conductive Seawater Background Suppression:** High seawater conductivity (3.0 to 5.2 S/m) generates rapid ambient eddy currents during active electromagnetic pulses; distinguishing the mineral deposit's signature requires microsecond-scale transient decay discrimination and **24-bit** dynamic range sampling.

- **Subsurface Drift & Georeferencing without GPS:** With satellite positioning blocked by the water column, cross-currents (0.05 to 0.3 m/s) cause unpowered pods to drift **100 to 500 meters** laterally during descent.

Low Cost Deployable Seafloor Metal Detection Sensor for Ocean Resource Exploration:

To design and develop a low cost deployable ocean bottom sensor that can be released from a research vessel during surveys to detect and map metal rich seabed deposits, including polymetallic nodules, hydrothermal sulphides, cobalt rich crusts and rare earth element bearing sediments, providing a rapid and cost effective tool for deep ocean mineral exploration.

1.4 Motivation

The ocean remains one of the least explored environments on Earth, making deep ocean research an exciting and challenging field of scientific investigation. This problem statement is particularly interesting from the perspective of physics students because it involves applying concepts from physics, mathematics, electronics, and engineering to a practical problem under extreme environmental constraints.

A major motivation for undertaking this project is the challenge of developing a functional deep-sea sensing system while keeping its cost low. Designing such a system requires us to apply our theoretical knowledge and analytical skills to problems involving sensing, signal detection, pressure, materials, electronics, and data analysis. This provides an opportunity to move beyond theoretical understanding and use our academic knowledge to develop a practical technological solution.

Through this project, we aim to demonstrate how fundamental scientific understanding and analytical thinking can be translated into an indigenous, real-world system that can contribute to ocean research and deep-sea resource exploration.

1.5 Objectives

The primary objectives of the proposed system are:

1. To design a compact and low cost deployable seafloor sensing payload.
2. To develop a sensing mechanism capable of detecting metallic signatures associated with metal rich seabed deposits.
3. To design a pressure-resistant and corrosion-resistant enclosure suitable for deep-ocean deployment.
4. To develop a data acquisition and processing system for recording and interpreting sensor measurements.

5. To develop a method for associating detected signals with geographical or spatial coordinates for seabed mapping.
6. To investigate a practical deployment and recovery mechanism that can be operated from a research vessel.
7. To validate the proposed system through laboratory testing and controlled underwater experiments.

2. System Design

2.1 Mechanical Design

The mechanical design of the proposed sensing module is driven primarily by the requirement to operate in the deep-ocean environment at a target depth of approximately 6000 m. At such depths, the enclosure is subjected to extremely high external hydrostatic pressure and must protect the electronic and sensing subsystems throughout deployment, operation, and recovery.

The proposed system is designed as a low-cost deployable sensing module rather than a self-propelled underwater vehicle. Multiple modules may be deployed as an array from a research vessel, allowed to descend to their respective measurement locations, used to acquire seabed data, and subsequently recovered. Since active underwater maneuverability is not a primary requirement, the mechanical design emphasizes pressure resistance, structural simplicity, low mass, manufacturability, and ease of recovery.

The mechanical architecture consists of a spherical outer pressure housing, a lightweight internal support structure, mounting provisions for the electronic subsystems, and a detachable ballast-based deployment and recovery mechanism. The present CAD model represents a preliminary design and will be refined following structural analysis, material testing, electronics integration, and experimental validation.

2.1.1 Design Requirements

The principal mechanical requirements of the proposed sensing module are summarized in Table 2.1.

Table 2.1: Primary mechanical design requirements.

Requirement	Design Consideration
Operating depth	Approximately 6000 m below the sea surface.
Pressure environment	High external hydrostatic pressure acting on the pressure housing.
Mission architecture	Multiple low-cost modules deployed as an array.
Underwater maneuverability	Not required for the present mission concept.
Internal payload	Custom PCB, sensors, battery, memory, and supporting electronics.
Sensor interface	Provision for external exposure or interface of selected sensors.
Descent	Detachable ballast used to provide negative buoyancy.
Recovery	Ballast release followed by buoyancy-assisted ascent.
Manufacturing	Preference for low-cost and indigenous manufacturing wherever practical.

2.1.2 Pressure Housing

The pressure housing forms the primary barrier between the internal electronic systems and the surrounding seawater. At the target depth, external hydrostatic pressure acts approximately normal to the surface of the enclosure and produces compressive loading throughout the housing. The design must therefore transfer the applied pressure through the shell while minimizing localized deformation and structural instability.

A spherical enclosure has been selected as the preliminary pressure housing because of its highly symmetric geometry and its suitability for external hydrostatic loading. The current concept consists of two hemispherical sections that can be assembled around the internal support structure and subsequently sealed to form the complete pressure housing.

The present CAD model is a preliminary configuration rather than a final pressure-rated design. The wall thickness, joining mechanism, sealing method, external openings, and detailed structural dimensions will be determined after analytical calculations, numerical simulation, material characterization, and experimental pressure testing.

The current engineering drawing represents an overall enclosure diameter of approximately 349.4 mm and a represented structural thickness of 20 mm. These dimensions are preliminary and may be modified during the detailed design stage.

Table 2.2: Preliminary dimensions of the spherical pressure housing.

Parameter	Preliminary Value
Overall diameter	349.4 mm
Represented structural thickness	20 mm
Configuration	Two hemispherical sections
Target operating depth	~ 6000 m

The dimensions presented in Table 2.2 correspond to the current CAD concept and are subject to optimization after structural analysis and system-level integration.

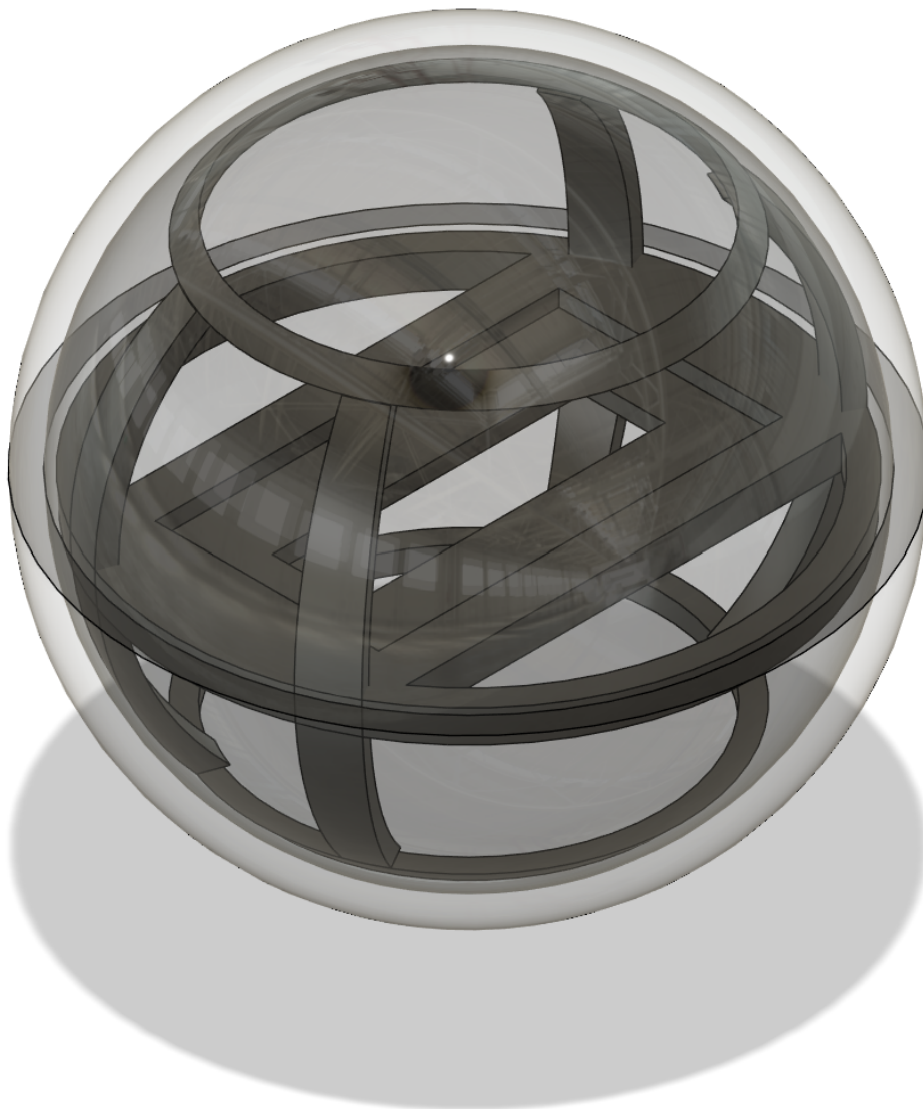


Figure 2.1: Preliminary engineering drawing of the spherical enclosure and central structural support.

The engineering drawing presents the preliminary geometry of the spherical enclosure and central support structure in multiple views, including top and side views. The current drawing is intended to communicate the initial mechanical envelope rather than represent

the final manufacturing drawing.

2.1.3 Geometry Selection

The selection of the pressure-housing geometry is primarily governed by its response to external hydrostatic pressure. A pressure-resistant shell should promote a stable and relatively uniform distribution of the applied load while minimizing localized deformation and the possibility of structural instability.

A spherical surface possesses curvature in two principal directions, providing a highly symmetric load-bearing configuration. In contrast, a cylindrical shell is curved in the circumferential direction but is essentially straight along its axial direction. This difference affects the distribution of membrane stresses and the susceptibility of the shell to buckling under external pressure.

The spherical geometry is also particularly suitable for the present mission architecture because the module is not intended to maneuver actively through the water column. Unlike an Autonomous Underwater Vehicle (AUV) or Remotely Operated Vehicle (ROV), the proposed module is designed to descend to the seabed, acquire measurements, and return to the surface after the mission. Consequently, hydrodynamic streamlining and active maneuverability are not primary design objectives at this stage.

The spherical design therefore provides a mechanically favorable configuration while allowing the overall system to remain simple and cost-effective.

A referenced finite-element study further illustrates the difference in linear buckling behaviour between spherical and cylindrical pressure vessels. The study reported substantially higher minimum buckling eigenvalues for comparable spherical configurations. For example, for a 15 mm Ti-10V-2Fe-3Al vessel, the reported eigenvalues were approximately 1399–1400 bar for the spherical configuration, compared with approximately 325–344 bar for the corresponding cylindrical configuration.

Table 2.3: Representative linear buckling eigenvalues for spherical and cylindrical pressure vessels reported in a referenced study.

Material	Geometry	Thickness (mm)	Eigenvalue 1 (bar)	Eigenvalue 3 (bar)
Ti-10V-2Fe-3Al	Sphere	15	1399	1400
Ti-10V-2Fe-3Al	Cylinder	15	325	344

The values in Table 2.3 are taken from a referenced finite-element study and are included only to illustrate the relative buckling behaviour of spherical and cylindrical geometries. They are not direct predictions for the proposed pressure housing because the material, dimensions, boundary conditions, and loading configuration are different. A dedicated structural and buckling analysis of the proposed enclosure is therefore required.

2.1.4 Sensor Housing

The sensor housing and mounting arrangement are designed to position the sensing elements relative to the seafloor while maintaining the required orientation during deployment. The exact arrangement will depend on the final sensor technology and its measurement geometry.

At least one sensing element may require direct interaction with the surrounding environment and may therefore need to be positioned outside the main pressure enclosure or connected through a dedicated external interface. The resulting opening and mounting arrangement will be treated as a critical part of the pressure-housing design.

Detailed sensor configuration and sensing physics are discussed separately in Chapter 5.

2.1.5 Material Selection

The material of the pressure housing is a critical design parameter and cannot be selected solely on the basis of mechanical strength. Since the proposed sensing system is intended to use magnetic measurements, the interaction between the enclosure material and the local magnetic environment must also be considered.

The principal material-selection criteria for the present design are summarized in Table 2.4.

Table 2.4: Material-selection criteria for the pressure housing.

Criterion	Design Requirement
Magnetic susceptibility	Low magnetic interaction to minimize disturbance of magnetic sensor measurements.
Hydrostatic pressure resistance	Adequate resistance to external pressure at the target operating depth.
Buckling resistance	Sufficient resistance against structural instability and collapse under external pressure.
Marine-environment compatibility	Adequate resistance to seawater and chemical exposure.
Availability in India	Practical accessibility and potential for domestic procurement.
Cost	Compatibility with the low-cost multi-module mission architecture.
Manufacturability	Ability to produce the required spherical geometry with consistent dimensional and surface quality.

Two candidate materials have been identified for further investigation: borosilicate glass and alumina ceramic.

Borosilicate glass is being considered because of its high compressive strength, chemical stability, transparency, and established use in scientific and underwater instrumentation. Its transparency may also facilitate visual inspection of the internal assembly. However, the availability of deep-sea-specific spherical pressure housings and the possibility of domestic customized manufacturing require further investigation.

Alumina ceramic is being considered because of its high hardness, compressive strength, chemical stability, and corrosion resistance. However, its suitability will depend on the achievable manufacturing quality, dimensional tolerances, fracture behaviour, joining method, and overall cost.

A preliminary comparison of the two candidate materials is given in Table 2.5.

Table 2.5: Preliminary comparison of candidate pressure-housing materials.

Property	Borosilicate Glass	Alumina Ceramic
Mechanical behaviour	High compressive strength; sensitive to surface defects	High hardness and compressive strength; brittle failure possible
Hydrostatic-pressure suitability	Requires analytical evaluation and pressure testing	Requires analytical evaluation and pressure testing
Magnetic interaction	Requires experimental evaluation	Requires experimental evaluation
Chemical stability	Good resistance to seawater and chemical attack	Excellent chemical stability and corrosion resistance
Transparency	Transparent	Generally opaque
Manufacturing	Specialized spherical glass fabrication required	Specialized ceramic manufacturing and finishing required
Availability in India	Deep-sea pressure-housing availability requires investigation	Suitable domestic manufacturing route requires investigation
Cost	To be evaluated	To be evaluated
Current status	Candidate material	Candidate material

The final material selection will therefore be based on a combined assessment of pressure resistance, buckling behaviour, magnetic susceptibility, fracture behaviour, density, manufacturability, availability in India, and cost. The final decision will be made only after numerical analysis and experimental testing.

2.1.6 Internal Support Structure

The spherical pressure housing contains a lightweight internal support structure that serves two primary purposes: supporting the internal electronic subsystems and maintaining their required orientation during deployment.

The current concept uses a modular rib-and-ring architecture consisting of upper and lower circular supports, a central PCB mounting support, and vertical structural members. This arrangement provides mounting locations for the electronic components while maintaining the relative geometry of the internal assembly.

The current engineering drawing identifies the major structural elements using the labels U, L, M, and R. The corresponding functions are listed in Table 2.6.

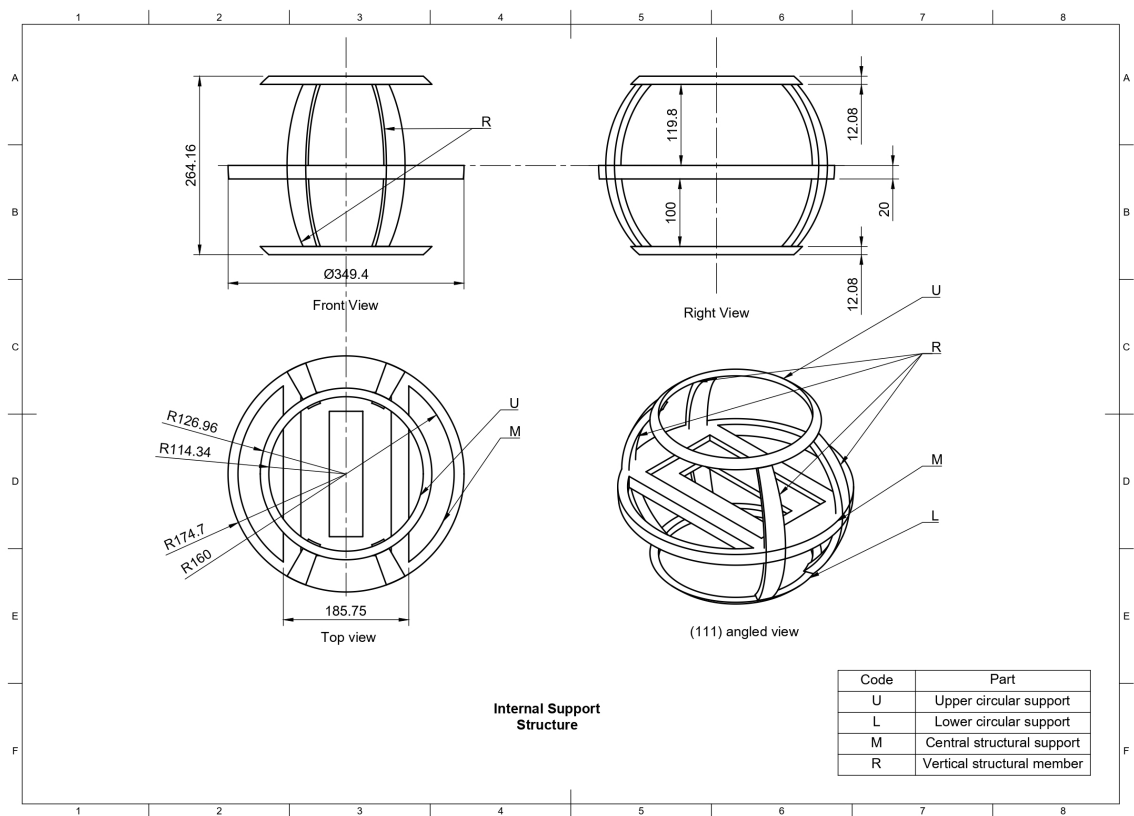


Figure 2.2: Preliminary engineering drawing of the internal support structure.

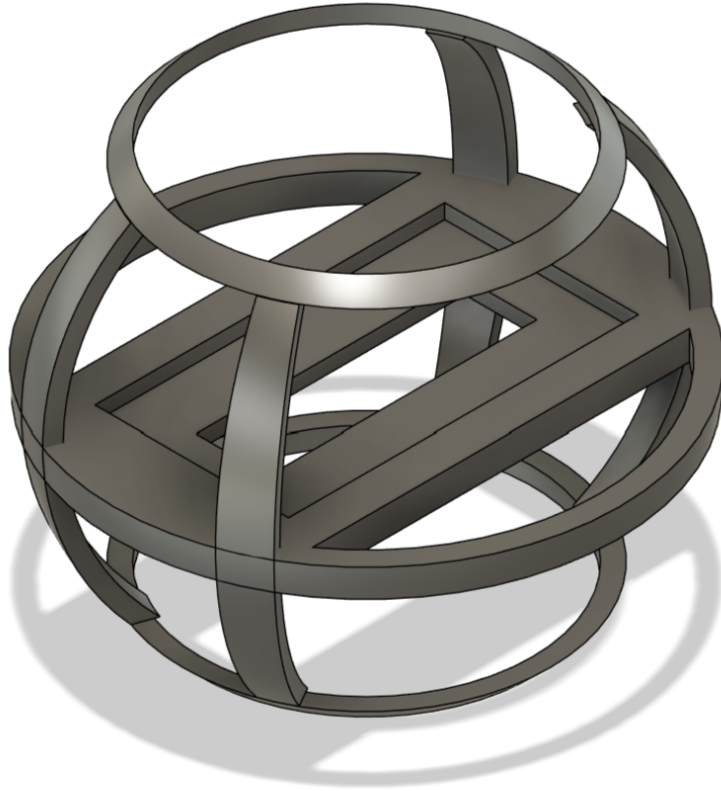


Figure 2.3: Internal arrangement of the spherical enclosure and structural support frame.

Table 2.6: Major components of the internal support structure.

Code	Component	Function
U	Upper circular support	Provides upper structural support and contributes to positioning of the internal assembly.
L	Lower circular support	Provides lower structural support and maintains the geometry of the internal frame.
M	Central structural support	Provides the primary mounting region for the custom PCB and associated electronics.
R	Vertical structural member	Connects the support rings and contributes to structural rigidity and orientation stability.

The current CAD configuration represents a central support width of approximately 185.75 mm and an overall internal structural height of approximately 264.16 mm. These dimensions are preliminary and will be modified according to the final dimensions of the custom PCB, battery, sensors, connectors, and other electronic subsystems.

The internal support structure is intended to be manufactured using additive manufacturing

for the prototype stage. PLA or another suitable lightweight polymer may be used initially, subject to evaluation of mechanical properties, environmental compatibility, and dimensional stability.

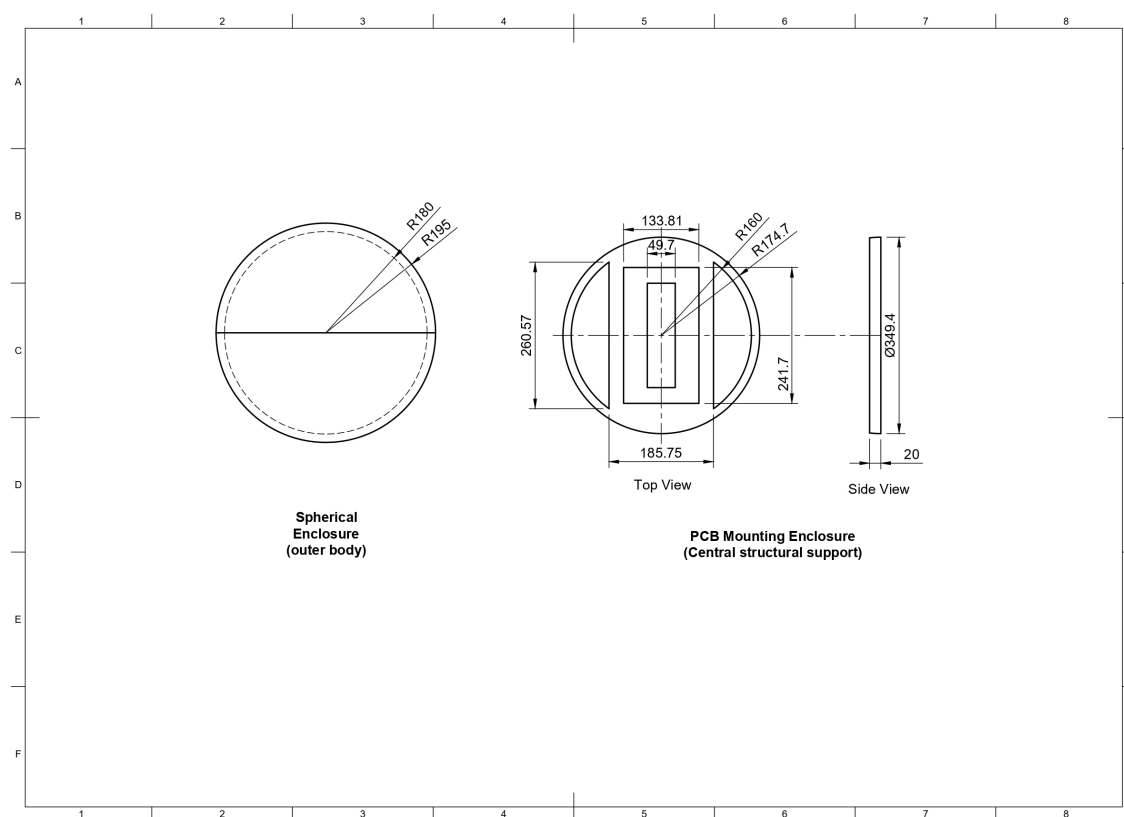
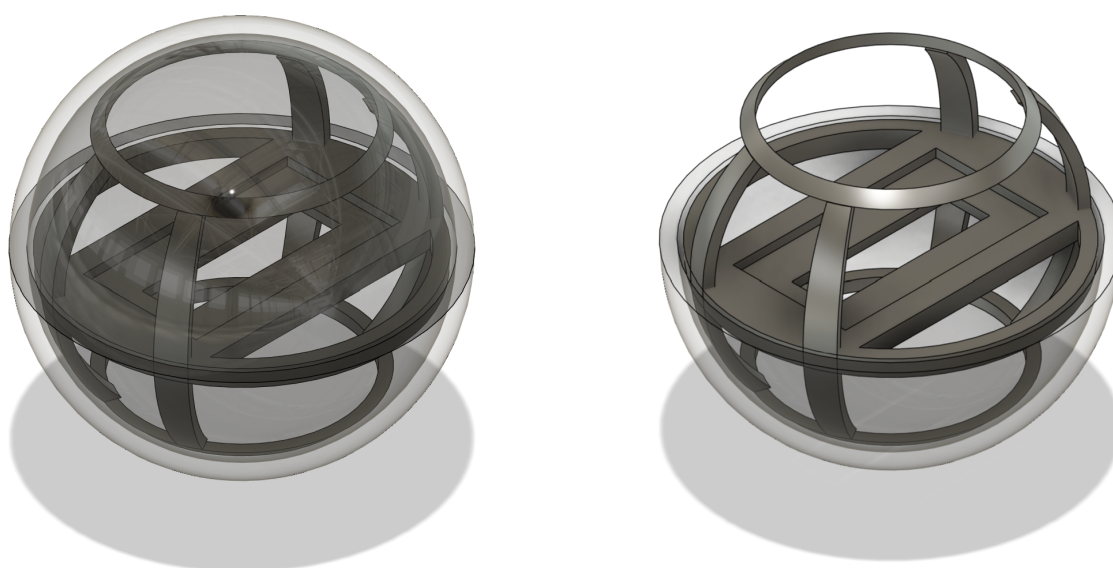


Figure 2.4: Preliminary engineering drawing of the spherical enclosure and internal support structure 1.



(a) Complete CAD assembly.

(b) Internal support structure.

Figure 2.5: Preliminary CAD design of the proposed spherical sensing module.

2.1.7 Sensor and Electronics Integration

The internal volume of the spherical enclosure will accommodate the sensing electronics, battery, data-storage system, and supporting circuitry. A custom PCB is planned to integrate the sensor electronics, signal-conditioning circuitry, power-management components, and relevant interfaces into a compact assembly.

The dimensions of the central mounting structure are therefore dependent on the final PCB and battery configuration. Since these dimensions are still under development, the present CAD model should be regarded as a preliminary mechanical envelope rather than the final internal assembly.

The internal frame will maintain the required orientation of the sensing system during deployment. Dedicated mounting locations will be provided for the PCB, battery, sensors, and associated components.

The final arrangement will also consider cable routing, connector placement, accessibility during assembly, sealing requirements, and thermal considerations.

2.1.8 Deployment and Recovery Mechanism

The proposed system is designed as a deployable sensing module rather than a self-propelled underwater vehicle. The low-cost mission architecture is based on deploying multiple modules as an array, allowing the modules to descend to the seabed, acquire measurements, and subsequently recover them.

A detachable ballast weight will be attached to the module during descent. The ballast provides the additional negative buoyancy required for the module to descend toward the target depth. Once the measurement operation is complete, the ballast will be released, allowing the positively buoyant module to ascend toward the surface.

This approach eliminates the need for onboard propulsion and active underwater maneuvering. Consequently, the mechanical system can remain comparatively simple, reducing the number of moving components, power requirements, and overall cost.

The final ballast-release mechanism and buoyancy configuration will be determined after the mass, displacement, and buoyancy characteristics of the completed module have been established.

Table 2.7: Proposed deployment and recovery sequence.

Stage	Operation
Deployment	Module is released from the research vessel with detachable ballast.
Descent	Ballast provides negative buoyancy and enables descent toward the target seafloor region.
Data acquisition	Sensors operate at the target depth and measurements are stored in onboard memory.
Ballast release	Ballast is detached after completion of the measurement operation.
Ascent	The remaining positive buoyancy causes the module to rise toward the surface.
Recovery	The module is recovered at the ocean surface and stored data are retrieved or transferred.

2.1.9 External Interfaces and Sensor Opening

The pressure housing cannot be treated as a completely uninterrupted sphere because certain sensing requirements may require an interface with the surrounding environment.

A dedicated opening will therefore be incorporated into the spherical enclosure for the sensor that requires external exposure or improved interaction with the surrounding seawater. The exact position of this opening, whether toward the upper or lower portion of the enclosure, will be determined according to the final sensing geometry and deployment orientation.

The opening represents a critical part of the pressure-housing design because it introduces a geometric discontinuity into the otherwise symmetric pressure boundary. Its dimensions, shape, sealing arrangement, and attachment mechanism will therefore require dedicated structural analysis and pressure testing.

The final enclosure may also require additional interfaces for charging, programming, assembly, or data retrieval. These interfaces will be designed to preserve pressure integrity and minimize unnecessary penetrations through the pressure boundary.

2.1.10 Manufacturability and Indigenous Development

The mechanical architecture is being developed with low-cost manufacturing and domestic production in mind. The internal support structure is intended to be produced using additive manufacturing, allowing rapid modification of the geometry as the PCB, battery, and sensor dimensions evolve.

The spherical pressure housing presents a greater manufacturing challenge and is therefore not expected to be fabricated using conventional desktop 3D printing. Instead, the enclosure will be sourced from specialized manufacturers capable of producing spherical glass or ceramic components with appropriate dimensional accuracy and surface quality.

Potential domestic manufacturing routes include companies serving the scientific-instrumentation,

laboratory-glassware, technical-ceramics, and specialized engineering sectors. The objective is to develop a design that can ultimately be manufactured and assembled within India wherever practical.

Indigenous manufacturing can reduce dependence on imported specialized pressure housings and may simplify customization, maintenance, future design modifications, and scaling to multiple sensing modules. It can also contribute to the development of domestic capability in underwater sensing, electronics, mechanical fabrication, materials, and ocean engineering. The preliminary manufacturing strategy is summarized in Table 2.8.

Table 2.8: Preliminary manufacturing strategy for major mechanical components.

Component		Proposed Manufacturing Method	Development Approach
Spherical housing	pressure	Specialized glass or ceramic fabrication	Domestic sourcing and customization where feasible
Internal support structure		Additive manufacturing	3D printing using PLA or another suitable lightweight polymer for prototype development
Custom sensor PCB		PCB fabrication and assembly	Indigenous design and system integration
Sensor interface		Machining or custom fabrication	Geometry finalized according to the selected sensor
Final assembly		In-house integration	Local assembly, testing, and validation

2.1.11 Design Validation and Testing

The current mechanical design is a preliminary concept and requires systematic validation before deployment in the deep ocean. Validation will be performed progressively, beginning with numerical analysis and laboratory testing and advancing toward controlled underwater experiments and eventual ocean deployment.

The first stage will involve computational analysis of the enclosure and internal support structure. Finite Element Analysis (FEA) will be used to investigate the structural response of the spherical housing under external hydrostatic pressure and to identify regions of high stress, deformation, and possible buckling. The influence of material properties, wall thickness, openings, joints, and support geometry will also be evaluated.

Following numerical analysis, a physical prototype will be manufactured and subjected to controlled laboratory testing. Initial tests will include sealing, buoyancy, orientation, mechanical integrity, and sensor operation. Pressure-chamber testing should subsequently be used wherever available to reproduce the pressure corresponding to the intended operating depth.

The physical testing programme will then progress toward deeper-water conditions before any full-scale ocean deployment is attempted.

Table 2.9: Planned mechanical validation pathway.

Stage	Method	Purpose
1	CAD modelling	Develop and optimize the enclosure and internal support geometry.
2	Numerical simulation	Evaluate stress, deformation, and buckling response under external pressure.
3	Prototype fabrication	Manufacture and assemble the initial physical prototype.
4	Leak and sealing tests	Verify sealing performance and basic pressure-housing integrity.
5	Pressure-chamber testing	Evaluate the enclosure under controlled hydrostatic pressure.
6	Shallow-water testing	Verify buoyancy, orientation, sensor operation, and recovery.
7	Deeper-water testing	Validate performance under progressively increasing environmental pressure and operational conditions.
8	Ocean testing	Evaluate the complete system under realistic deployment conditions.

The proposed development pathway can therefore be summarized as:

CAD Design → Numerical Simulation → Prototype Fabrication → Leak/Sealing Tests → Pressure-Chamber Testing → Shallow-Water Testing → Deeper-Water Testing → Ocean Testing

The final deep-ocean configuration will only be considered after the intermediate validation stages demonstrate adequate structural integrity, pressure resistance, buoyancy behaviour, sensor functionality, and recovery performance.

2.2 Electronic Design

The electronic subsystem is designed around a multimodal sensing architecture combining magnetic, acoustic, and controlled-source electromagnetic (CSEM) measurements. These primary sensing modalities are supported by navigation and environmental sensors that provide information about platform motion, orientation, depth, and temperature.

The purpose of combining multiple sensing modalities is to identify spatial anomalies associated with metal-rich seabed regions rather than relying on a single sensor for direct elemental identification. The processed measurements from the different sensors will subsequently be combined to generate a mineral-prospectivity or anomaly map.

The overall electronic architecture is intended to remain compact and low-power so that

it can operate within the pressure-protected enclosure for the duration of the underwater mission.

2.2.1 Sensors

The proposed sensing system employs three primary sensing modalities: magnetic sensing, acoustic sensing, and controlled-source electromagnetic (CSEM) sensing. In addition, an internal inertial measurement unit (IMU), a pressure sensor, and a temperature sensor are used for navigation, orientation correction, depth estimation, environmental monitoring, and data synchronization.

The primary sensors and their roles are summarized below.

Table 2.10: Primary sensor modalities and their intended functions.

Sensor	Measurement	Primary Function
Magnetic sensor	Local magnetic field and field gradient	Detection of magnetic anomalies associated with metal-rich seabed regions.
Acoustic sensor	Seafloor acoustic backscatter	Seabed profiling and identification of changes in seafloor or nodule coverage.
CSEM sensor	Seafloor electrical response	Detection of conductivity contrasts associated with conductive mineralized structures.
IMU	Acceleration and angular velocity	Orientation, motion estimation, and correction of orientation-dependent measurements.
Pressure sensor	External hydrostatic pressure	Depth estimation and pressure monitoring.
Temperature sensor	Temperature	Environmental monitoring and temperature-dependent measurement correction.

Magnetic Sensor

At least two spatially separated three-axis magnetic sensors are proposed. The sensors will be positioned at separated locations on the sensor body to establish a measurement baseline and enable estimation of the local magnetic-field gradient.

For the prototype stage, a magneto-inductive sensor such as the PNI RM3100 is considered because of its compact size and relatively low power consumption. For the final deep-sea target system, a low-noise fluxgate sensor such as the SENSYS FGM3D UW is preferred. Alternative fluxgate sensors may also be evaluated during the development process.

The final magnetic sensor will be selected after considering cost, pressure-compatible pack-

aging, calibration requirements, environmental compatibility, and experimentally measured noise performance.

Table 2.11: Preliminary comparison of candidate magnetic sensors.

Parameter	PNI RM3100	SENSYS FGM3D UW	Bartington Mag639	Stefan Mayer FL1-100	Prototype	Target System
Technology	Magneto-inductive	3-axis fluxgate	3-axis fluxgate	Fluxgate	–	–
Axes	3-axis	3-axis	3-axis	1-axis	–	–
Range	$\pm 1100 \mu\text{T}$	$\pm 100 \mu\text{T}$	$\pm 100 \mu\text{T}$	$\pm 100 \mu\text{T}$	–	–
Noise	$\approx 15 \text{ nT}$	$< 15 \text{ pT}/\sqrt{\text{Hz}} @ 1 \text{ Hz}$	$< 20 \text{ pT}/\sqrt{\text{Hz}} @ 1 \text{ Hz}$	$< 20 \text{ pT}/\sqrt{\text{Hz}} @ 1 \text{ Hz}$	–	–
Bandwidth	Up to 147 Hz	DC–2 kHz	DC–12 kHz	DC–1 kHz	–	–
Output	SPI / I ² C	Analog	Analog	Analog	–	–
Deep-sea suitability	Requires pressure housing	Designed for underwater use	Requires pressure protection	Requires pressure protection	–	–
Proposed use	Prototype	Preferred final candidate	Alternative	Alternative	–	–

At least two spatially separated magnetic sensors enable calculation of the differential magnetic field,

$$\Delta \mathbf{B} = \mathbf{B}_2 - \mathbf{B}_1 \quad (2.1)$$

where $\mathbf{B}_1$ and $\mathbf{B}_2$ are the magnetic-field vectors measured by the two sensors.

This differential measurement provides sensitivity to local magnetic-field gradients and can reduce the influence of magnetic variations that are approximately common to both sensors. It does not, however, completely eliminate environmental or platform-generated magnetic fields. Therefore, the sensors will be positioned away from batteries, power converters, electromagnetic actuators, and other potential sources of magnetic interference. Calibration and experimental characterization will be required to determine the achievable measurement accuracy.

Acoustic Sensor

Acoustic sensing is based primarily on the measurement of acoustic backscatter from the seafloor. A short acoustic pulse is transmitted toward the seabed and the returned signal is recorded by the receiver. Changes in the returned acoustic response can provide information about seafloor structure and variations in nodule coverage.

The principal acoustic quantities considered by the system are:

- **Time of Flight (ToF):** determines the acoustic propagation distance between the sensor and the seafloor and supports seabed profiling.
- **Acoustic Power Loss:** represents the reduction in transmitted acoustic power during propagation and reception.
- **Acoustic Pressure:** represents the pressure fluctuation associated with the returning acoustic wave.
- **Backscatter Strength:** represents the strength of the acoustic energy returned from the seabed and can be used to identify variations in seabed or nodule coverage.

The acoustic power-loss relation is expressed as

$$TL = 10 \log_{10} \left(\frac{P_{TX}}{P_{RX}} \right), \quad (2.2)$$

where P_{TX} is the transmitted acoustic power and P_{RX} is the received acoustic power.

A polyurethane (PU) acoustic interface or window is proposed for the acoustic path because its acoustic impedance can be made closer to that of seawater than that of a conventional rigid pressure-hull material. This can reduce acoustic reflection at the water–window interface.

A PZT piezoelectric transducer is proposed behind the acoustic interface for acoustic transmission and reception. The acoustic interface is not treated as the primary pressure boundary; the pressure-resistant housing must continue to protect the internal electronics.

Controlled-Source Electromagnetic (CSEM) Sensor

The controlled-source electromagnetic (CSEM) subsystem uses a time-varying current through an electromagnetic transmitter coil to generate a controlled primary electromagnetic field. This field interacts with the conductive seawater and seafloor.

Conductive mineralized structures, particularly seafloor massive sulphides (SMS), can produce secondary electromagnetic responses. These responses can be detected using seawater-contacting receiver electrodes and/or magnetic receivers.

The CSEM subsystem consists of the following principal components:

- **EM transmitter coil:** generates the controlled primary electromagnetic field.
- **Receiver electrodes:** measure the voltage difference associated with the electric field in seawater.
- **Signal-conditioning circuit:** amplifies and filters the weak received signal.
- **Analog-to-digital converter (ADC):** converts the conditioned analogue signal into digital data.
- **Microcontroller (MCU):** controls the excitation sequence, records synchronized measurements, and performs signal processing.

The electric field can be estimated from the measured electrode voltage difference as

$$E = \frac{\Delta V}{L}, \quad (2.3)$$

where E is the electric field in V/m, ΔV is the measured voltage difference in V, and L is the separation between the receiver electrodes in m.

Navigation and Environmental Sensors

In addition to the primary detection sensors, the system incorporates navigation and environmental sensors to provide contextual information and to enable correction of measurement data.

An internal inertial measurement unit (IMU) measures acceleration and angular velocity to estimate platform motion and orientation. IMU measurements are used to correct orientation-dependent magnetic, acoustic, and electromagnetic measurements.

A pressure transducer measures the external hydrostatic pressure and provides an estimate of deployment depth. A temperature sensor records the internal or local system temperature and can be used for monitoring and subsequent measurement correction.

Inertial Measurement Unit (IMU) A compact six-axis MEMS IMU is proposed for the prototype system. Candidate devices include the TDK ICM-42605, Bosch BMI270, and TDK ICM-42688-P.

Table 2.12: Preliminary comparison of candidate IMUs.

Parameter	TDK ICM-42605	Bosch BMI270	TDK ICM-42688-P
Technology	6-axis MEMS IMU	6-axis MEMS IMU	6-axis MEMS IMU
Axes	3-axis accelerometer + 3-axis gyroscope	3-axis accelerometer + 3-axis gyroscope	3-axis accelerometer + 3-axis gyroscope
Accelerometer range	$\pm 2/4/8/16\ g$	$\pm 2/4/8/16\ g$	$\pm 2/4/8/16\ g$
Gyroscope range	Up to $\pm 2000^\circ/\text{s}$	Up to $\pm 2000^\circ/\text{s}$	Up to $\pm 2000^\circ/\text{s}$
Interface	SPI / I ² C / I ³ C	SPI / I ² C	SPI / I ² C / I ³ C
Proposed use	Preferred	Alternative	High-performance alternative

Temperature Sensor A digital temperature sensor is included for environmental monitoring and temperature-dependent measurement correction. The preliminary candidates are TI TMP117, TI TMP102, and DS18B20.

Table 2.13: Preliminary comparison of candidate temperature sensors.

Parameter	TI TMP117	TI TMP102	DS18B20
Type	Digital temperature sensor	Digital temperature sensor	Digital temperature sensor
Temperature range	-55 to $+150^\circ\text{C}$	-40 to $+125^\circ\text{C}$	-55 to $+125^\circ\text{C}$
Accuracy	Up to $\pm 0.1^\circ\text{C}$ typical	Up to $\pm 0.5^\circ\text{C}$	Up to $\pm 0.5^\circ\text{C}$
Interface	I ² C	I ² C	1-Wire
Proposed use	Preferred	Alternative	Low-cost alternative

Pressure Sensor The pressure sensor provides an estimate of the deployment depth from external hydrostatic pressure. Different pressure sensors are considered for the prototype and final system because the pressure requirement at 6000 m is substantially greater than that of shallow-water testing.

Table 2.14: Preliminary pressure-sensor options for different development stages.

Parameter	Deep-Sea Pressure Transducer	TE MS5837-30BA	TE MS5837-02BA	Proposed Use
Type	External hydrostatic pressure sensor	Digital pressure sensor	Digital pressure sensor	–
Pressure range	≥ 600 bar target	0–30 bar	0–2 bar	–
Approx. depth capability	≈ 6000 m	≈ 300 m	≈ 20 m	–
Interface	Analog / digital depending on model	I ² C	I ² C	–
Proposed use	Final system	Prototype testing	Shallow-water testing	–

The pressure-sensor selection will ultimately depend on the required pressure range, accuracy, environmental compatibility, packaging, and availability.

2.2.2 Microcontroller

The microcontroller acts as the central control and synchronization unit for the electronic subsystem. It will coordinate the primary sensing systems, acquire measurements from the different sensors, record time-synchronized data, and control the excitation sequence of the CSEM subsystem.

The microcontroller will also interface with the IMU, pressure sensor, temperature sensor, and data-storage system. The final device selection will be based on the required number of interfaces, ADC performance, processing capability, memory requirements, power consumption, and compatibility with the custom PCB.

Since the detailed sensor-fusion algorithm and final data-processing requirements are still under development, the specific microcontroller will be finalized after the sensor-interface architecture and power budget have been established.

2.2.3 Power System

The power subsystem will provide regulated electrical power to the microcontroller, sensors, signal-conditioning circuits, communication system, data-storage system, and other electronic components.

Since the sensing module is intended for autonomous operation during its descent, seafloor measurement, and ascent, the power system must support the complete mission duration while remaining sufficiently compact for integration inside the pressure housing.

The final battery capacity will be determined from the measured power consumption of the

individual subsystems and the expected mission duration. Power-management circuitry will be incorporated into the custom PCB to provide regulated supply voltages and appropriate protection for the electronic components.

Low-power operation will be prioritized through appropriate duty cycling, sleep modes, and controlled activation of high-power subsystems where possible. This is particularly relevant to the CSEM transmitter and acoustic subsystem, which may not require continuous operation.

A detailed power budget will be presented in Section 7.3 after the individual component selections and operating modes have been finalized.

2.2.4 Data Storage and Synchronization

The sensing module will include onboard non-volatile memory for local storage of sensor measurements during the underwater mission. Since continuous high-bandwidth communication through the water column is not assumed, local storage provides a reliable means of preserving the complete measurement dataset for post-mission analysis.

The microcontroller will time-synchronize the measurements obtained from the magnetic, acoustic, CSEM, IMU, pressure, and temperature subsystems. Platform orientation, pressure/depth, and temperature will therefore be recorded together with the primary sensor measurements.

This synchronized dataset will allow the measurements obtained from different sensing modalities to be associated during subsequent processing and spatial mapping.

2.2.5 Preliminary Electronic Integration

The final electronic system is intended to be implemented on a custom PCB to reduce wiring complexity and provide a compact, organized integration of the sensor interfaces, signal-conditioning circuits, microcontroller, power-management circuitry, and data-storage system.

The magnetic sensors will be placed away from power electronics and other sources of magnetic interference. The acoustic transducer will be positioned with a clear acoustic path toward the seabed, while the CSEM transmitter and receiver electrodes will be positioned according to the required electromagnetic measurement geometry.

The IMU will remain inside the pressure-protected electronics compartment, while the pressure sensor will interface with the external environment through an appropriate pressure port. The final PCB dimensions and component arrangement will be determined after the individual component selection and mechanical packaging are finalized.

The resulting electronic architecture will therefore be developed jointly with the mechanical enclosure and sensor configuration rather than as an independent subsystem.

2.3 Software Design

The software subsystem is responsible for acquiring, synchronizing, preprocessing, storing, and interpreting the measurements obtained from the multimodal sensor system. The proposed software architecture is designed to combine magnetic, acoustic, controlled-source electromagnetic (CSEM), pressure, temperature, and inertial measurement data into a unified dataset for subsequent anomaly detection and mineral prospectivity mapping.

The machine-learning component is intended to assist in the interpretation of multimodal seafloor measurements. Rather than assuming that any individual sensor can directly identify the elemental composition of a deposit, the system converts synchronized measurements into feature vectors representing measurable characteristics of the seabed. Machine-learning models can then be used to identify relationships between these features and mineral-associated conditions and to highlight regions requiring further investigation and ground-truth sampling.

2.3.1 Software Architecture

The proposed software pipeline consists of five principal stages:

1. **Data acquisition:** Synchronized measurements are collected from the magnetic, acoustic, CSEM, pressure, temperature, and IMU subsystems.
2. **Preprocessing:** Raw sensor streams are organized, calibrated, filtered, and converted into suitable feature representations. Pressure, temperature, and IMU measurements are used to provide environmental and motion-related corrections.
3. **Feature generation:** Relevant characteristics are extracted from the processed measurements to construct feature vectors representing the measured seafloor conditions.
4. **Machine-learning inference:** Suitable candidate machine-learning models are trained and evaluated using appropriate experimental datasets and are subsequently used to identify anomalous or mineral-associated measurement patterns.
5. **Output generation and validation:** Model predictions are converted into a mineral-prospectivity or anomaly map and validated against controlled test data and, subsequently, field measurements.

The overall processing sequence can therefore be represented as:

Sensor Data → Preprocessing → Feature Extraction → ML Model → Anomaly Detection → Prospectivity Mapping

The machine-learning stage is intended to complement, rather than replace, the underlying physical sensing methods.

2.3.2 Multimodal Data Representation

The proposed system produces measurements from multiple sensing modalities. These measurements are synchronized using the microcontroller and supplemented with pressure, temperature, and orientation information.

The resulting measurements are represented as feature vectors that provide a measurable “fingerprint” of the local seabed conditions.

A representative feature vector can be expressed conceptually as

$$\mathbf{x} = [\mathbf{x}_M, \mathbf{x}_A, \mathbf{x}_E, \mathbf{x}_P, \mathbf{x}_T, \mathbf{x}_I], \quad (2.4)$$

where:

- $\mathbf{x}_M$ represents features derived from magnetic measurements;
- $\mathbf{x}_A$ represents acoustic features;
- $\mathbf{x}_E$ represents controlled-source electromagnetic features;
- $\mathbf{x}_P$ represents pressure or depth information;
- $\mathbf{x}_T$ represents temperature information; and
- $\mathbf{x}_I$ represents motion and orientation information obtained from the IMU.

The exact composition of the feature vector will be determined experimentally after characterization of the individual sensing channels.

2.3.3 Machine Learning Strategy

AquaMiner currently considers multiple machine-learning approaches rather than committing to a single model at the proposal stage. The candidate methods have different computational and data requirements and are therefore intended for evaluation using the experimental dataset generated during system development.

The candidate methods and their intended roles are summarized in Table 2.15.

Table 2.15: Candidate machine-learning methods and their proposed roles in AquaMiner.

ML Method	Proposed Role
YOLOv7-based Model	Candidate deep-learning approach for detecting mineral-related targets in suitable sensor or imaging data, depending on the availability of labeled training data.
XGBoost	Tree-based boosting method for learning relationships between synchronized sensor features and mineral-associated conditions.
Support Vector Machine (SVM)	Supervised-learning method for classifying feature vectors into different mineral-associated conditions.
Convolutional Neural Network (CNN)	Deep-learning method for extracting spatial or local patterns from suitable structured sensor data or imagery.
Random Forest (RF)	Ensemble tree-based method for classification and analysis of multimodal sensor features.
Gaussian Process Regression (GPR)	Probabilistic regression method for estimating continuous mineral-related responses and associated uncertainty when suitable training data are available.
U-Net	Image-segmentation architecture for identifying spatial regions of interest in suitable acoustic or other sensor-derived images.

The candidate methods differ substantially in their requirements and intended applications. Tree-based methods such as XGBoost and Random Forest may be suitable for structured multimodal feature vectors, whereas CNN- and U-Net-based approaches are more naturally suited to spatially structured data or imagery. SVM can provide a classification approach for relatively well-defined feature spaces, while Gaussian Process Regression can provide both predictions and associated uncertainty when an appropriate regression dataset is available.

A YOLOv7-based model is considered only for applications in which the available sensing data can be represented in a suitable target-detection format and sufficient labeled training data can be generated.

2.3.4 Model Selection

The final machine-learning model will not be selected solely on the basis of theoretical capability. Model selection will depend on the nature and quantity of experimentally acquired data, the availability and quality of labels, computational requirements, validation performance, and the ability of the model to generalize to previously unseen seabed conditions.

The selection procedure will therefore involve comparison of candidate models using a common experimental dataset and appropriate performance metrics. The final model may also differ for different sensing modalities if the underlying data characteristics require separate approaches.

At the current stage, all methods listed in Table 2.15 should therefore be considered candidate approaches rather than final design selections.

2.3.5 Machine Learning Workflow

The proposed machine-learning workflow consists of data acquisition, preprocessing, model development, output generation, and validation.

Table 2.16: Proposed machine-learning processing workflow.

Stage	Description
Data acquisition	Collect synchronized magnetic, acoustic, CSEM, pressure, temperature, and IMU measurements.
Preprocessing	Calibrate, filter, synchronize, and organize the sensor streams while applying appropriate environmental and motion corrections.
Feature generation	Extract relevant measurable features and construct feature vectors representing local seabed conditions.
Model development	Train and evaluate suitable candidate machine-learning models using labeled or otherwise appropriate experimental datasets.
Output generation	Convert model predictions into anomaly or mineral-prospectivity information for subsequent mapping.
Validation	Validate individual sensing channels before integrating the complete multimodal system, followed by system-level testing.

2.3.6 Output and Mineral Prospectivity Mapping

The final output of the software subsystem is intended to be a spatial representation of anomalous or potentially mineral-associated regions of the seabed.

Model predictions from the multimodal sensor data will be associated with the corresponding depth, orientation, and positional information. These results can then be used to generate a preliminary mineral-prospectivity map indicating regions that may warrant detailed investigation.

The prospectivity output is intended as a decision-support product rather than a direct determination of elemental composition. Regions identified as anomalous can subsequently be prioritized for detailed surveys, ground-truth sampling, or other high-resolution investigation.

2.3.7 Machine Learning Validation

Validation will be performed progressively, beginning with individual sensor channels and controlled laboratory targets before integration of the complete multimodal system.

The first stage will evaluate the repeatability and noise characteristics of the individual sensors. Controlled targets representing different metal-associated conditions will then be

used to generate experimental datasets for model development.

Candidate models will be compared using appropriate validation procedures and performance metrics. The selected approach will subsequently be tested on measurement conditions not used during training to assess generalization.

Finally, the multimodal machine-learning system will be evaluated as an integrated pipeline using synchronized measurements from the complete sensor platform.

The final detection threshold, feature set, and sensor-fusion strategy will therefore be established experimentally rather than fixed a priori.

3. Sensor and Detection Methodology

3.1 Signal Processing

Raw magnetic data $\rightarrow$ calibration $\rightarrow$ orientation correction $\rightarrow$ gradient

Raw acoustic data $\rightarrow$ pulse detection $\rightarrow$ ToF $\rightarrow$ backscatter features

Raw CSEM data $\rightarrow$ synchronization $\rightarrow$ background estimation $\rightarrow$ response features

IMU/pressure/temperature $\rightarrow$ motion/depth/environmental corrections

4. Deployment Strategy

4.1 Depth Requirement

Discuss the target operating depth.

For example:

$$P = P_0 + \rho gh \tag{4.1}$$

where

- P is the pressure at depth h ,
- P_0 is the atmospheric pressure,
- ρ is the density of seawater,
- g is the acceleration due to gravity,
- h is the depth.

5. Design Calculations

5.0.1 Comparison of Geometries

A referenced finite-element study comparing spherical and cylindrical pressure vessels reported substantially higher linear buckling eigenvalues for spherical configurations under comparable conditions. For example, for a 15 mm Ti-10V-2Fe-3Al vessel, the reported first three eigenvalues were approximately 1399–1400 bar for the spherical configuration, compared with approximately 325–344 bar for the corresponding cylindrical configuration [?].

These results provide supporting evidence for selecting a spherical geometry for the present concept. However, the reported values are not directly applicable to the proposed enclosure because the material, dimensions, boundary conditions, and loading configuration differ. A dedicated buckling analysis of the proposed housing will therefore be required.

Discuss the structural requirements of the enclosure.

5.1 Power Budget

Table 5.1: Estimated power consumption of major components.

Component	Power (W)	Quantity
Sensor	—	—
Microcontroller	—	—
Communication System	—	—
Other Electronics	—	—

6. Cost Analysis

Table 6.1: Estimated project cost.

Component	Quantity	Estimated Cost (INR)
Sensor	1	—
Microcontroller	1	—
Battery	1	—
Pressure Housing	1	—
Electronics	—	—
Miscellaneous	—	—
Total		—

7. Expected Results and Impact

The proposed system is intended to provide a low-cost and deployable platform for preliminary deep-ocean mineral exploration. Since the project is currently at the design and prototype-development stage, the performance values presented in this chapter represent expected outcomes rather than experimentally validated results.

The system is expected to combine magnetic, acoustic, and controlled-source electromagnetic measurements with environmental and motion data to identify regions exhibiting signatures associated with metal-rich seabed conditions. The resulting measurements can subsequently be used to support preliminary target identification, seabed mapping, environmental observation, and the planning of more detailed surveys.

7.1 Expected Performance

The proposed sensing module is expected to operate as a compact, recoverable seabed-sensing platform capable of acquiring synchronized measurements during a deep-ocean deployment.

The expected performance of the system includes:

1. Reliable acquisition of magnetic, acoustic, and controlled-source electromagnetic measurements from the seafloor environment.
2. Simultaneous recording of pressure, temperature, orientation, and motion information to provide contextual information for the primary sensor measurements.
3. Local storage of the acquired measurements throughout the deployment and recovery cycle.
4. Identification of spatial anomalies or changes in seabed characteristics that may indicate regions of increased mineral prospectivity.
5. Generation of a preliminary spatial representation of the measured seabed characteristics for subsequent analysis and mapping.
6. Reusability of the sensing module through recovery after completion of the measurement operation.

The final performance of the system will depend on the selected sensors, signal-conditioning electronics, mechanical configuration, deployment conditions, and the results obtained from experimental validation.

7.2 Expected Detection Range

The effective detection range of the proposed system cannot be specified reliably before the sensor configuration and measurement methodology have been experimentally characterized. The achievable range will depend on several factors, including sensor sensitivity, target size and composition, sensor-to-seabed distance, orientation, environmental noise, seawater conductivity, and the signal-processing technique employed.

Rather than defining a fixed detection distance at this stage, the project will experimentally determine the relationship between target properties, sensor response, and separation distance.

For magnetic sensing, the use of spatially separated sensors is expected to allow measurement of local magnetic-field differences and gradients. For acoustic sensing, the returned signal will provide information regarding seafloor structure and backscatter characteristics. The CSEM subsystem is expected to provide complementary information related to electrical conductivity contrasts associated with conductive mineralized structures.

The final detection range, sensitivity, minimum detectable target, and measurement uncertainty will therefore be established through controlled laboratory and underwater experiments using representative targets.

7.3 Advantages

The proposed architecture has several advantages over a system based entirely on a large mobile underwater platform.

7.3.1 Low-Cost Architecture

The system is designed around relatively compact sensing modules rather than a single highly complex underwater vehicle. This approach can reduce the mechanical, propulsion, control, and operational requirements associated with a fully mobile AUV or ROV.

7.3.2 Modular Deployment

Multiple modules can potentially be deployed during the same survey to collect measurements from different seabed locations. A modular architecture also allows individual sensing units to be recovered, inspected, repaired, or modified without replacing an entire survey platform.

7.3.3 Reduced Mechanical Complexity

The proposed module does not require continuous underwater propulsion or active navigation. Its primary functions are descent, measurement, recovery, and data storage. Eliminating unnecessary maneuvering hardware reduces system complexity and power requirements.

7.3.4 Multimodal Sensing

The combination of magnetic, acoustic, and electromagnetic measurements provides complementary information about seabed conditions. This reduces the dependence on a single physical measurement and provides a broader basis for identifying regions of potential interest.

7.3.5 Indigenous Development

The project emphasizes the development of an indigenous sensing platform that can be designed, integrated, tested, and progressively manufactured within India. This can facilitate customization for Indian oceanographic applications and reduce dependence on imported specialized instrumentation.

7.4 Environmental Benefits

The proposed system is intended primarily for sensing and observation rather than direct extraction of seabed material. Its stationary or hovering operation and non-invasive multimodal sensing approach can reduce the need for repeated movement of heavy equipment, excavation, or unnecessary physical sampling during preliminary surveys.

A sensing-first approach may therefore help narrow down regions of interest before more invasive or resource-intensive survey activities are undertaken. This can potentially reduce unnecessary seabed disturbance and limit repeated sampling in regions with low mineral prospectivity.

The measurements collected by the system may also contribute to environmental baseline studies by providing geophysical and environmental information about the seabed. Such measurements can support the identification of potentially sensitive regions and improve the planning of future surveys, monitoring activities, and ground-truth sampling.

The environmental benefits described in this section are potential outcomes of the proposed architecture. Their actual magnitude will require validation through field measurements and comparison with conventional survey practices.

7.5 Economic Benefits

A major objective of the proposed system is to reduce the cost and operational complexity associated with preliminary deep-ocean surveys.

By separating the sensing function from a fully mobile AUV or ROV platform, the proposed architecture may reduce the equipment and operational requirements needed for preliminary target identification. The use of multiple relatively simple modules may also provide greater survey flexibility while avoiding the need to rely on a single complex platform for every measurement.

The modular design may further reduce maintenance costs because individual sensing components or modules can be inspected, repaired, upgraded, or replaced independently.

By identifying potentially promising regions before detailed sampling, the system may also help reduce unnecessary drilling, physical sampling, survey time, and associated operational expenditure in low-prospectivity areas.

The expected economic advantage therefore arises not only from the individual component cost, but also from the overall mission architecture, deployment strategy, reusability, and reduction of unnecessary survey operations.

7.6 National Impact

The proposed system is particularly relevant to India's growing interest in deep-ocean exploration and mineral-resource assessment in the Indian Ocean.

India has allocated an exploration area of approximately 75,000 km² in the Central Indian Ocean Basin for polymetallic nodule exploration, together with additional exploration areas associated with polymetallic sulphides. These resources may contain metals of strategic interest, including nickel, copper, cobalt, manganese, and other economically important elements.

The proposed platform is intended to complement such exploration activities by providing a low-cost means of collecting multiparameter seabed data. Instead of replacing existing research vessels, AUVs, ROVs, or other advanced systems, the proposed technology can serve as a complementary preliminary-survey tool.

The project also aligns with India's broader Deep Ocean Mission, which includes deep-sea mining, underwater robotics, deep-ocean surveys, biodiversity-related activities, and ocean-resource development.

A major national objective of the project is indigenous technological development. Designing and integrating the sensing, electronics, mechanical structure, data system, and supporting hardware within India can contribute to domestic capability in underwater instrumentation, ocean engineering, electronics, materials, and scientific sensing.

The present system is therefore intended not only as a student-developed prototype but also as a potential foundation for further development by academic institutions, research laboratories, and Indian ocean-technology organizations.

7.7 Scientific Benefits

A principal scientific advantage of the proposed system is its ability to collect multiple types of measurements from the same region of the seafloor.

The proposed sensing architecture combines:

- magnetic-field measurements;
- acoustic backscatter and seabed-profile information;
- controlled-source electromagnetic measurements;
- pressure and depth measurements;
- temperature measurements; and
- platform motion and orientation measurements.

The combination of these measurements can provide a more comprehensive representation of the local seabed environment than any single sensing modality alone.

The processed measurements can be combined to generate a preliminary anomaly or mineral-prospectivity indicator. Such an indicator is intended to identify regions that warrant further investigation rather than to directly determine the exact elemental composition of a deposit.

The system may therefore support several scientific activities, including preliminary mineral exploration, seabed characterization, environmental baseline observations, and the planning of subsequent high-resolution surveys and ground-truth sampling.

An additional scientific benefit is the opportunity to develop and validate a low-cost multi-parameter sensing methodology specifically suited to deep-ocean deployment. The resulting datasets may also be useful for studying the relationship between different geophysical signatures and seabed conditions.

7.8 Summary of Expected Impact

The expected impact of the proposed system can be summarized across four major dimensions: environmental, economic, national, and scientific.

Table 7.1: Summary of expected impact of the proposed sensing platform.

Impact Area	Expected Contribution
Environmental	Non-invasive sensing may reduce unnecessary seabed disturbance and support environmental baseline studies.
Economic	Low-cost modular deployment may reduce equipment and operational requirements for preliminary surveys.
National	Supports India's deep-ocean exploration objectives and development of indigenous underwater technology.
Scientific	Provides synchronized magnetic, acoustic, electromagnetic, environmental, and motion measurements for seabed characterization.
Technological	Encourages indigenous development in underwater sensing, electronics, mechanical design, materials, and data processing.

The actual performance, operational depth, detection accuracy, and environmental or economic benefits of the proposed system will ultimately depend on experimental validation. The expected outcomes presented here therefore serve as the design targets and potential impact of the proposed technology rather than as experimentally established results.

8. Future Scope

Discuss possible improvements and future extensions.

Possible future developments include:

- Autonomous navigation.
- Real-time communication.
- AI-assisted detection.
- Improved sensor sensitivity.
- Integration with an underwater remotely operated vehicle.